

In-season adaptations to intense intermittent training and sprint interval training in sub-elite football players

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This study investigated the in-season effect of intensified training comparing the efficacy of duration-matched intense intermittent exercise training with sprint interval training in increasing intermittent running performance, sprint ability, and muscle content of proteins related to ion handling and metabolism in football players. After the first two weeks in the season, 22 sub-elite football players completed either 10 weeks of intense intermittent training using the 10-20-30 training concept (10-20-30, $n = 12$) or sprint interval training (SIT, $n = 10$; work/rest ratio: 6-s/54-s) three times weekly, with a ~20% reduction in weekly training time. Before and after the intervention, players performed a Yo-Yo intermittent recovery test level 1 (Yo-Yo IR1) and a 30-m sprint test. Furthermore, players had a muscle biopsy taken from the *vastus lateralis*. Yo-Yo IR1 performance increased by 330 m (95%CI: 178-482, $P \leq 0.01$) in 10-20-30, whereas no change was observed in SIT. Sprint time did not change in 10-20-30 but decreased by 0.04 second (95%CI: 0.00-0.09, $P \leq 0.05$) in SIT. Muscle content of HADHA (24%, $P \leq 0.01$), PDH-E1 α (40%, $P \leq 0.01$), complex I-V of the electron transport chain (ETC) (51%, $P \leq 0.01$) and Na⁺, K⁺-ATPase subunits α_2 (33%, $P \leq 0.05$) and β_1 (27%, $P \leq 0.05$) increased in 10-20-30, whereas content of DHPR (27%, $P \leq 0.01$) and complex I-V of the ETC (31%, $P \leq 0.05$) increased in SIT. Intense intermittent training, combining short sprints and a high aerobic load, is superior to regular sprint interval training in increasing intense intermittent running performance during a Yo-Yo IR1 test and muscle content of PDH-E1 α and HADHA in sub-elite football players.

KEYWORDS

HIIT, HIT, oxidative metabolism, SIT, Soccer, speed endurance training

1 | INTRODUCTION

The ability to perform and recover from high-intensity runs and sprints is important in intermittent team sports such as football.¹⁻³ The most pronounced differences in match activity pattern between professional and amateur players are the distance and number of high-intensity runs and sprints performed during the first and second half.¹ Over the past decades, a variety of training strategies have been investigated

to optimize physical performance of football players.⁴⁻⁶ An effective strategy to enhance football-specific performance is intense intermittent exercise training performed at supramaximal intensity (ie, running velocities above that corresponding to maximal oxygen consumption ($\dot{V}O_{2max}$)).⁷⁻⁹

In active individuals, improvements in performance are closely related to the training modality undertaken,^{10,11} suggesting that training specificity is important. Sub-elite football players, training 2-3 times per week, have limited time

available to train the technical, tactical, and physical aspects required in football. Given the complex nature of football requiring players to possess both speed and conditioning, it is relevant to investigate whether sub-elite football players benefit from integrating low-volume intense intermittent exercise combining sprint efforts with high aerobic load, as this may be time efficient to allow time for training with a technical and tactical focus. Studies show that intense intermittent training comprising of long-duration sprints (~30-second) with relatively long recovery periods (work:rest ratio of ~1:6) improves intermittent exercise capacity.^{7,8} However, in trained football players, speed endurance training does not necessarily enhance sprint performance.¹² On the other hand, shorter duration all-out sprints (≤ 10 seconds), conducted in conjunction with a high aerobic load, may improve intermittent exercise capacity, but it is unknown whether such training regime also improves sprint performance, as expected by traditional sprint training with short all-out sprints (~6 seconds) with long recovery periods. A type of intense intermittent training that combines 10-second sprints with a high aerobic load is the 10-20-30 training concept.¹³ While this type of training enhances aerobic endurance capacity,¹³⁻¹⁶ it is unknown whether it also improves sprint performance.

Although studies have elaborated on the effectiveness of intense intermittent training¹⁷ and sprint training on performance in football players,^{18,19} the mechanisms underlying performance enhancements remain poorly understood.⁸ In a recent study, speed endurance training, consisting of 6-10 $\times$ 30-second all-out runs with 3-minute recovery, improved shuttle run performance and repeated sprint ability, as well as increased abundance of muscle proteins regulating ion transport and oxidative metabolism.²⁰ In addition, muscle content of Na⁺, K⁺-ATPase subunits and maximal activity of oxidative enzyme 3-hydroxyacyl-CoA-dehydrogenase (HADHA) were shown to predict the capacity to perform high-intensity runs in football players.²¹ Sprint training, consisting of 6-10-second duration sprints, on the other hand, improves acceleration and sprint ability in trained football players,⁹ which are associated with higher anaerobic enzyme activity and abundance of muscle ion transport proteins involved in Ca²⁺ and H⁺ handling.^{10,22} Repeated short (~5 seconds) sprints with short recovery periods (~10 seconds) may enhance intense intermittent running performance during a Yo-Yo intermittent recovery test level 1 (Yo-Yo IR1), despite a decrease in muscle content of proteins involved in oxidative metabolism in well-trained football players,⁵ implying that further investigations are required to understand skeletal muscle adaptations underpinning training-induced improvements in football-specific performance.

In the present study, we investigated the in-season effect of 10 weeks of duration-matched intense intermittent training and short-duration sprint interval training on intermittent running and sprint performance, as well as skeletal muscle

characteristics in ion handling capacity and oxidative metabolism in sub-elite football players. We hypothesized that intense intermittent training, combining all-out sprint efforts with a high aerobic load, would be superior to traditional sprint interval training in enhancing intermittent running performance in sub-elite football players.

2 | METHODS

2.1 | Subjects

The study was performed in a team squad of 25 competitive sub-elite football players (aged 19-34 years) playing in the second best amateur league in Denmark. Players were healthy and did not use any prescription medicine. The study was approved by the local ethics committee of Copenhagen and conducted in accordance with the Helsinki-II-Declaration. Oral and written informed consent was obtained from all subjects before inclusion.

2.2 | Study design

The study was designed as a randomized parallel trial that compared the in-season effect of 10 weeks of intense intermittent training with sprint interval training. The primary outcome measure was change in intense intermittent running performance during a Yo-Yo IR1 test, which has been shown to be sensitive to measure changes in football-specific performance and to be related to high-intensity runs during matches.^{23,24}

The study was initiated two weeks in-season (April), following a pre-season starting late January. During the second week in-season, players' intense intermittent running performance and sprint ability were determined at two separate days before and after the 10-week training intervention. Day 1 and day 2 were conducted 52 and 100 hours after a match, respectively, both before and after the intervention. Order of days was the same before and after the intervention, and players were instructed not to engage in physical activity 24 hours before test days. To determine intense intermittent running performance, players performed a Yo-Yo IR1 test, whereas players' sprint ability was tested with a 30-m straight line sprint test and a change of direction agility test (*T* test)²⁵ following a standardized warm up. In addition, players had a biopsy (~100 mg) taken from the *vastus lateralis* muscle during local anesthesia (lidocaine without epinephrine, Xylocain 20 mg $\times$ mL⁻¹, AstraZeneca) using a Bergström needle with suction in a rested state 52 hours after the first and last match of the season corresponding to 7 days before the first and 5 days after the last day of testing, respectively. Biopsies were analyzed for fiber type composition and content of proteins involved in ion handling and lactate transport as well as in glycolytic and oxidative metabolism.

2.3 | Training intervention

After testing, players were randomized, stratified for Yo-Yo IR1 performance, to an intense intermittent training group using the 10-20-30 training concept (10-20-30) or a sprint interval training group (SIT) for a 10-week intervention. Before the intervention, the team had a weekly training frequency of three sessions per week with a training time of ~1:40 h:min per session. During the 10-week intervention, training frequency was unchanged, but weekly time spent during training was reduced by 20 minutes (~20%) per training session in both groups. The intervention replaced moderate intensity small sided games and set pieces (eg, corners and free kicks). The last 14-23 minutes of each training session was devoted to the specific training regimen for each group. In 10-20-30, players performed 2-3 sets of 5 repetitions of 30 seconds of jogging, 20 seconds at a moderate speed, and 10 seconds at maximal sprint effort.¹³ This type of training was shown to enhance 1.5 and 5 km running performance in trained runners by 6% and 4% after 7 weeks of training.¹³ Players conducted 2 sets of 10-20-30 in weeks 1-4 and 3 sets in weeks 5-10, with 4 minutes of recovery between each set. In SIT, players performed 2-3 sets of 6 change-of-direction sprints (45°-90°) at maximal effort for 6-second, each interspersed by 54-second of passive recovery. Despite a lower metabolic load during change-of-direction sprints (45°-90°) compared to straight line sprints,²⁶ change-of-direction sprints were chosen because of a higher relevance toward match play. Players conducted 2 sets in weeks 1-4 and 3 sets in weeks 5-10 with 4 minutes recovery between each set. Total time spent during the two specific training regimes were matched between the two groups (14 minutes during weeks 1-4 and 23 minutes during weeks 5-10). The rationale for choosing these training regimens was to compare all-out exercise protocols of similar duration and short intense efforts at maximal intensity but with different muscle metabolic response. Specifically, more severe muscle metabolic and ionic perturbations are expected during 10-20-30 training. Pilot data retrieved from 4 trained male subjects before the study revealed peak mixed venous blood lactate concentrations of 10-23 mmol/L and K⁺ concentrations of 5.5-6.2 mmol/L (ABL flex 800, Radiometer, Copenhagen, Denmark) during one training session of 10-20-30. Such values are higher than those normally reported during repeated short sprints interspersed with relatively long (exercise:rest ratio >1:8) recovery periods.^{10,27} In contrast, the SIT protocol was chosen as a traditional sprint training protocol aiming at improving speed (ie, peak speed and the ability to accelerate and decelerate).

2.4 | Myosin heavy chain isoform distribution

Myosin heavy chain (MHC) composition of the *vastus lateralis* muscle was determined from homogenate using gel electrophoresis as described previously.²⁸ Muscle homogenate

(80 µL) was mixed with 200 µL of sample buffer (10% glycerol, 5% 2-mercaptoethanol and 2.3% SDS, 62.5 mmol/L Tris and 0.2% bromophenol blue at pH 6.8), boiled in water at 100°C for 3 minutes and loaded (10-40 µL) on a SDS-PAGE gel (6% polyacrylamide (100:1 acrylamid: bis-acrylamid), 30% glycerol, 67.5 mmol/L tris-base, 0.4% SDS, and 0.1 M glycine). Gels were run at 80 V for at least 42 hours at 4°C. MHC bands were visualized (ChemiDoc MP Imaging System) using stain-free imaging technology and were quantified densitometrically using imaging software (Image Lab v. 4.0, Bio-Rad Laboratories, Hercules, CA, USA).

2.5 | Muscle protein content

Protein content in muscle homogenate was determined by Western blotting as previously described.²⁹ In short, ~2 mg of dry weight (dw) muscle tissue was dissected free from connective tissue and homogenized on ice in a fresh buffer (10% glycerol, 20 mmol/L Na-pyrophosphate, 150 mmol/L NaCl, 50 mmol/L HEPES, 1% Nonidet P-40, 20 mmol/L β-glycerophosphate, 10 mmol/L NaF, 2 mmol/L PMSF, 1 mmol/L EDTA, 1 mmol/L EGTA, 10 g × mL⁻¹ aprotinin, 10 g × mL⁻¹ leupeptin, and 3 mmol/L benzamidine) with a Polytron 3100 (Kinematica) for 30 seconds. Samples were rotated end-over-end for 1 hours at 4°C before being centrifuged for 30 minutes at 17 500 g at 4°C. The supernatant was collected and analyzed for protein concentration by ELISA using BSA standards (Pierce Reagents). For Western blotting, lysates were diluted in 6 × sample buffer (0.5 M Tris-base, DTT, SDS, glycerol, and bromophenol blue) to a final protein concentration of 2 µg × mL⁻¹ and loaded in 26-well TGX Criterion gels (4%-15% Tris-HCL; Bio-Rad Laboratories, Hercules, CA, USA). Samples from the same subject were loaded next to each other on the same gel. Proteins were separated by SDS-PAGE for ~60 minutes at 250 V. After SDS-PAGE, gels were transferred (TransBlotTurbo, Bio-Rad Laboratories, Hercules, CA, USA) to a polyvinylidene difluoride (PVDF) membrane (Transfer Pack, Bio-Rad Laboratories, Hercules, CA, USA) for 7.5-10 minutes at 2.5 mA and 20-25 V. Membranes were blocked in Tris-buffered saline including 0.1% Tween-20 (TBST) with 2% milk for 30 minutes and then incubated in primary antibodies overnight at 5°C. Primary antibodies used are presented in Table 1. After two washes in TBST, membranes were incubated for 1 hour in horseradish peroxidase-conjugated secondary antibody (DAKO, Copenhagen, Denmark) diluted 1:5000 in TBST with 2% milk. Membranes were then washed in TBST, after which bands were visualized using chemiluminescent detection (Super Signal West Femto Maximum Sensitivity Substrate; ThermoScientific, Rockford, IL, USA) and images were collected on the ChemiDoc MP Imaging System. Band intensities were quantified using imaging software (Image Lab v. 4.0, Bio-Rad Laboratories, Hercules, CA, USA). One

TABLE 1 Antibodies used for immunoblotting

Antibody	Reference	Manufacturer	Source	Dilution	Apparent MW (kDa)
DHPR	ab2864	Abcam	Mouse	1:500 (2% milk)	170
HADHA	ab54477	Abcam	Rabbit	1:1000 (2% milk)	85
MCT 4	AB3316P	Merck Millipore	Rabbit	1:1000 (3% BSA)	50
Na ⁺ /K ⁺ -ATPase subunit α_2	07-674	Merck Millipore	Rabbit	1:500 (2% milk)	100
Na ⁺ /K ⁺ -ATPase subunit β_1	MA3-930	Thermo Fisher Scientific Inc	Mouse	1:1000 (2% milk)	50
ETC complex I-V	MS601	MitoSciences	Mouse	1:500 (2% milk)	18-54
PDH-E1 α	sc-377092	Santa Cruz Biotechnology	Mouse	1:3000 (2% milk)	43
PFK	sc-166722	Santa Cruz Biotechnology	Mouse	1:2000 (3% BSA)	85

MW, molecular weight.

sample was excluded from further analysis because of very low and unquantifiable signals (from SIT).

2.6 | Statistics

Statistical analyses were performed in SPSS version 24 (IBM, Armonk, NY, USA). Data were tested for normality using the Shapiro Wilk's test and Q-Q plots. Data are presented as means and standard deviations (SD). To estimate between-group differences in training-induced changes in outcome measures, a linear mixed model was used with group as a fixed factor. Performance-related outcome measures were tested both with and without inclusion of baseline value as a covariate in the model.³⁰ Outcome statistics are presented as means with 95% confidence intervals (CI) unless otherwise specified and *P*-values to represent probability. Inference-based statistics was used to assess the magnitude of an effect for performance-related measures using Cohen's standardized mean difference (Cohen's *d*), where 0.2-0.5 is a small effect, 0.5-0.8 a moderate effect, and >0.8 a large effect. The level of significance was defined as $P \leq 0.05$.

3 | RESULTS

3.1 | Subjects characteristics, training compliance, and injuries

Of the team squad of 25 players, 22 were included in the study. Two players from SIT did not complete the study because of match-related injuries during the intervention. Characteristics of the 20 players who completed the study are presented in Table 2. Training compliance was 82 (± 10)% in 10-20-30 and 83 (± 8)% in SIT with no difference between the groups. Body mass did not change in either group during the intervention.

3.2 | Performance measures

Outcome statistics for Yo-Yo IR1 performance, change of direction *T* test and 30-m sprint performance with and without inclusion of baseline covariate are presented in Table 3. Baseline scores had little impact on performance-related outcome measures. In 10-20-30, a moderate enhancement of 18% was observed in Yo-Yo IR1 performance with the intervention, whereas no change was observed in SIT (Table 3, Figure 1). 10-20-30 had superior improvements in Yo-Yo IR1 performance than SIT. In 10-20-30, no changes were observed in change of direction *T* test and 30-m sprint time with the intervention, whereas in SIT, a small enhancement of 1% was observed in both *T* test and 30-m sprint performance (Table 3). The difference of change in *T* test and 30-m sprint performance between 10-20-30, and SIT was significant in favor of SIT.

3.3 | Muscle MHC-isoform distribution

No changes were observed in distribution of MHCI and MHCII isoforms with the intervention in either group (Figure 2). No apparent bands were detectable for MHCIIx.

TABLE 2 Subject characteristics of the intense intermittent training group (10-20-30) and sprint interval training group (SIT)

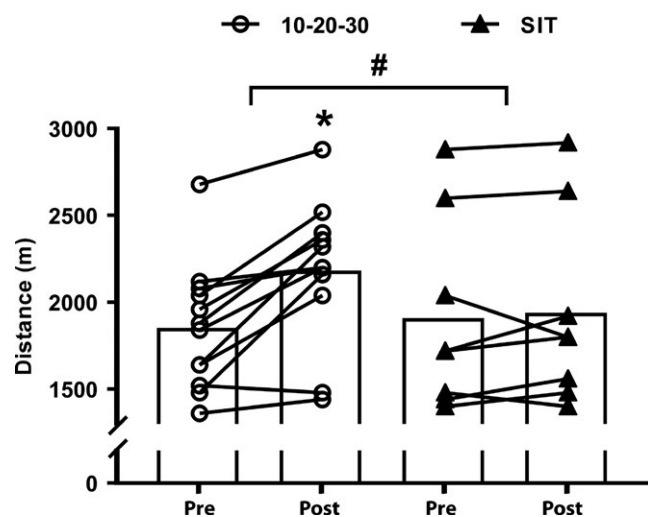
	10-20-30 (n = 12)	SIT (n = 8)
Age (years)	22.8 (± 2.2)	24.9 (± 5.4)
Height (m)	1.78 (± 0.05)	1.81 (± 0.05)
Body mass (kg)	75.1 (± 6.4)	76.5 (± 8.0)
BMI ($\text{kg} \times \text{m}^{-2}$) ⁻¹	23.5 (± 1.8)	23.3 (± 1.5)

Values are mean ($\pm$ SD).

BMI, body mass index.

TABLE 3 Parameter estimates for performance measures before and after 10 wk intense intermittent training (10-20-30) and sprint interval training (SIT)

	Mean values ($\pm$ SD)		Between-group comparisons			
	10-20-30 (n = 12)	SIT (n = 8)	Group difference (95% CI)	P-value	d	Effect
Yo-Yo IR1 distance covered (m)						
Baseline	1853 ($\pm$ 362)	1910 ($\pm$ 557)				
Follow-up	2183 ($\pm$ 401)	1940 ($\pm$ 553)				
Change score	330 ($\pm$ 239)	30 ($\pm$ 136)	300 (123-477)	0.002	0.69	Moderate
Change score with baseline covariate	328 ($\pm$ 249)	31 ($\pm$ 144)	296 (109-484)	0.004	0.68	Moderate
30 m sprint time (s)						
Baseline	4.31 ($\pm$ 0.12)	4.34 ($\pm$ 0.16)				
Follow-up	4.32 ($\pm$ 0.07)	4.30 ($\pm$ 0.12)				
Change score	0.01 ($\pm$ 0.06)	-0.04 ($\pm$ 0.05)	0.05 (0.00-0.10)	0.055	0.38	Small
Change score with baseline covariate	0.00 ($\pm$ 0.03)	-0.04 ($\pm$ 0.04)	0.04 (0.03-0.08)	0.036	0.31	Small
T test sprint time (s)						
Baseline	9.32 ($\pm$ 0.16)	9.34 ($\pm$ 0.23)				
Follow-up	9.33 ($\pm$ 0.13)	9.29 ($\pm$ 0.28)				
Change score	0.01 ($\pm$ 0.05)	-0.05 ($\pm$ 0.07)	0.06 (0.00-0.13)	0.053	0.32	Small
Change score with baseline covariate	0.01 ($\pm$ 0.03)	-0.05 ($\pm$ 0.06)	0.06 (0.01-0.12)	0.029	0.32	Small

**FIGURE 1** Distance covered in the Yo-Yo intermittent recovery level 1 test before (PRE) and after (POST) 10 weeks of intense intermittent training (Open circles, 10-20-30: n = 12) and sprint interval training (Filled triangles, SIT: n = 8) in sub-elite football players. Individual values and bars indicate group mean. #Group $\times$ trial interaction ($P \leq 0.05$). *Different ($P \leq 0.05$) from Pre

3.4 | Muscle ion handling and metabolic proteins

Muscle content of DHPR tended ($P = 0.07$) to increase by 11% with the intervention in 10-20-30, whereas content

increased ($P \leq 0.01$) by 27% in SIT, with no between-group difference (Figure 2). In 10-20-30, content of Na^+ , K^+ ATPase subunits α_2 and β_1 increased by 33% and 27% ($P \leq 0.05$), respectively, with the intervention, whereas no changes were observed in SIT (Figure 2). In 10-20-30, content of HADHA and PDH-E1 α was 24% and 40% higher ($P \leq 0.01$), respectively, after the intervention compared with before, whereas no change was observed in SIT (Figure 2). In both 10-20-30 ($P \leq 0.01$) and SIT ($P \leq 0.05$) mean content of subunits of the electron transport chain (ETC) complex I-V increased with the intervention, whereas no changes were observed in content of PFK in either group (Figure 3). Training-induced increases in content of HADHA and PDH-E1 α were greater in 10-20-30 than SIT (Figure 3).

4 | DISCUSSION

The novel findings of the present study are that intense intermittent training, combining short sprints and a high aerobic load, is superior to regular sprint interval training in increasing intense intermittent running performance during a Yo-Yo IR1 test and muscle protein content of PDH-E1 α and HADHA in sub-elite football players. On the other hand, sprint interval training improved 30-m sprint and T test agility

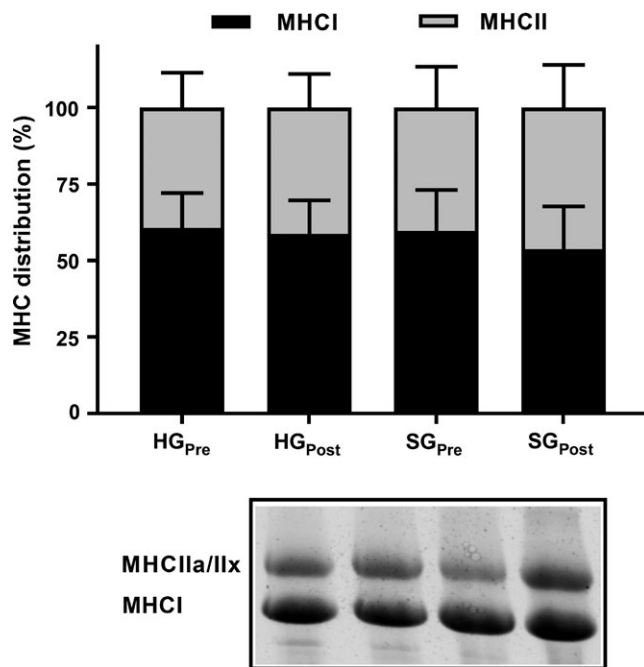


FIGURE 2 Myosin heavy chain (MHC) isoform distribution in the vastus lateralis muscle before (PRE) and after (POST) 10 weeks of intense intermittent training (10-20-30: $n = 12$) or sprint interval training (SIT: $n = 8$) in sub-elite football players. Values are mean and SD

performance. Enhancement in Yo-Yo IR1 performance was associated with an increase in muscle proteins involved in ion handling and oxidative metabolism.

In spite of a 20% reduction in weekly training time, we observed beneficial, but different, performance-related adaptations when duration-matched intense intermittent training or sprint training was added within the regular training among sub-elite football players. While intense intermittent training was superior to sprint training in enhancing Yo-Yo IR1 performance, sprint training was more effective in improving change of direction *T* test and 30-m sprint performance, implying that training characterized by different intensity, duration, and exercise:recovery ratio promotes different adaptations. The importance of using appropriate exercise: recovery ratio in augmenting training-induced enhancements in intense intermittent running performance has been highlighted by Iaia et al, showing greater enhancement in Yo-Yo IR2 performance in young professional football players when 6-8 $\times$ 20-second all-out sprints were interspersed by 120 vs 40 seconds of recovery.^{4,9}

Intensified training with a concomitant reduction in training time enhances intense intermittent exercise performance in football players,^{18,19} cyclists,³¹ and runners.^{32,33} Studies comparing intense intermittent training, differing in intensity and/or exercise and rest duration, show that training affects

football-specific performance in a modality-dependent manner.^{4,11} However, discrepancies exist, in that 2 weeks of intensified training consisting of 10-12 $\times$ 30 seconds sprints with 3-minute recovery enhanced repeated sprint ability but not Yo-Yo IR2 performance in sub-elite football players.^{18,19} This discrepancy may be related to the relatively short intervention of only two weeks in players already accustomed to intense training or that the Yo-Yo IR2 has higher variability than the Yo-Yo IR1 (CV: 9.6% vs 4.9%).^{24,34} Thus, it cannot be excluded that players in the latter studies improved their intermittent endurance performance. Nonetheless, it is expected that individuals with lower fitness will be more prone to enhance performance in response to intense intermittent training.

In the present study, we chose 10-20-30 training, as this type of intense intermittent training improves cardiorespiratory fitness and running performance, while also including short duration all-out sprints.¹³ Accordingly, one of the mechanisms underlying enhancements in performance with 10-20-30 training seems to be an increase in $\dot{V}O_{2\max}$.¹³ Although we did not measure $\dot{V}O_{2\max}$ in the present study due to time constraints in season, we did observe some indications of oxidative adaptations in skeletal muscle. The 10-20-30 training increased muscle protein content of oxidative enzymes PDH-E1 α and HADHA, as well as content of mitochondrial ETC complexes, with 10-20-30 training being superior to sprint training in upregulating protein content of PDH-E1 α and HADHA. This suggests that the greater enhancement in Yo-Yo IR1 performance induced by 10-20-30 training may be explained by a greater energy flux toward aerobic metabolism and/or by a postponed reliance on anaerobic metabolism, thereby delaying fatigue development during the test. A further mechanism underlying the greater improvement in Yo-Yo IR1 performance may be related to improved ion regulation as indicated by an increase in muscle content of Na⁺, K⁺-ATPase subunits α_2 and β_1 . Greater abundance of muscle Na⁺, K⁺-ATPase subunits incurred by high-intensity intermittent training is associated with improved K⁺ handling during and in recovery from intense exercise and with enhanced performance.^{8,31} In support of these putative mechanisms, content of HADHA and Na⁺, K⁺-ATPase subunits has been shown to be important determinants of running capacity and high-intensity performance, respectively, in football players.²¹ In addition, given no changes were observed in sprint and agility performance in 10-20-30, it is unlikely the enhancement in Yo-Yo IR1 performance was related to an increased ability to accelerate or perform high-speed turns. This is further supported by the improved sprint performance observed in SIT without concomitant increases in Yo-Yo IR1 performance.

Apart from enhanced sprint performance in SIT, we observed that muscle content of the voltage-dependent calcium channel DHPR increased with the intervention. This

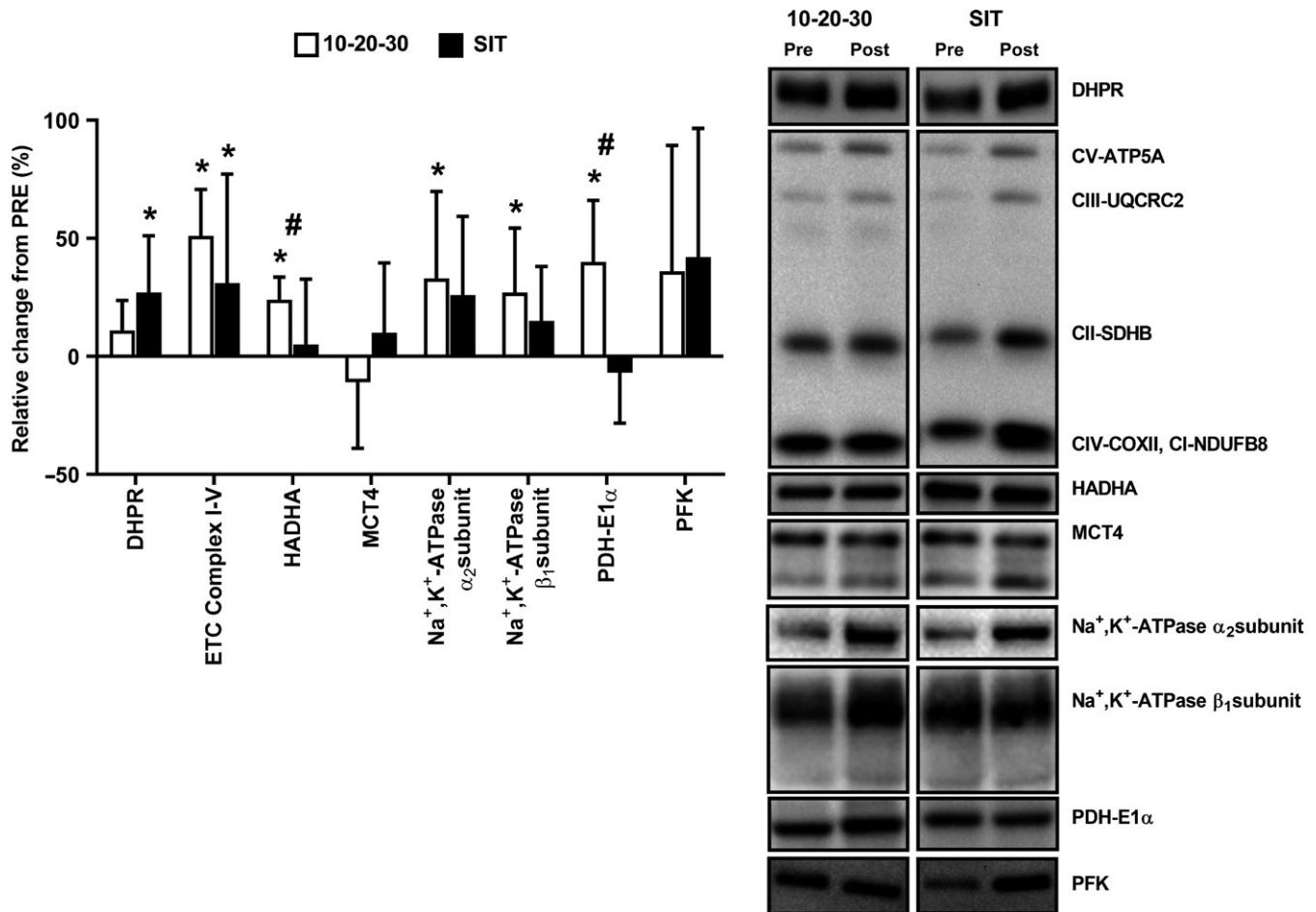


FIGURE 3 Muscle content of protein related to ion handling, lactate transport, and metabolism before (PRE) and after (POST) 10 weeks of intense intermittent training (Open bars, 10-20-30; $n = 12$) or sprint interval training (Filled bars, SIT; $n = 7$) in sub-elite football players. Values are mean and upper bound of the 95% CI. #Group $\times$ trial interaction ($P \leq 0.05$). *Different ($P \leq 0.05$) from PRE. Representative blots are shown on the right side of the figure

observation is in line with the sarcoplasmic reticulum expansion and enhanced Ca^{2+} release induced by sprint interval training observed in trained cyclists.²² An improved muscle Ca^{2+} handling could potentially be beneficial for muscle force production and thus sprint performance.^{22,35} Indeed, animal studies show that DHPR is an important determinant of sprint velocity.³⁵ Nonetheless, it may be that the enhancement of sprint performance in SIT is related to adaptations not investigated in the present study, including changes within the central nervous system, tendons, and muscle hypertrophy.

The increase in muscle proteins related to Ca^{2+} and K^{+} handling in SIT (ie, DHPR) and 10-20-30 (ie, Na^{+} , K^{+} -ATPase subunits α_2 and β_1) coincides with another study comparing adaptations to two different intense intermittent training regimes.¹⁰ However, that study included recreationally active subjects, and no study has to our knowledge compared muscle adaptations to different intense intermittent training modalities in trained football players in season. Fransson et al observed that 6-10 $\times$ 30-second sprint intervals for 4 weeks improved Yo-Yo IR2 performance more than small-sided

games conducted as 2 $\times$ 7 minutes, despite no differences in muscle adaptations between the exercise modalities.²⁰ In addition, in well-trained football players, 9 weeks of intensified training performed in-season enhanced Yo-Yo IR1 performance despite no changes in content of muscle mitochondrial enzymes and capillarization,⁵ suggesting a complex relationship between intense intermittent performance and adaptations to intensified training performed in-season in trained football players.

The metabolic and ionic perturbations elicited during 10-20-30 training are expected to be different from short-duration sprint interval training based on accumulation of blood metabolites and heart rate response.^{10,13,27} Specifically, 10-20-30 training elicits peak blood lactate levels of 10-23 mmol/L and an average heart rate response of $\sim 85\%$ of maximum heart rate, with $\sim 45\%$ of total training time spent above 90% of maximum.¹³ In comparison, 6-second all-out sprints, interspersed by $\frac{1}{2}$ -1 minute recovery may yield blood lactate levels of 5-10 mmol/L and an average heart rate response of 70%-75% of maximum.^{10,27} Thus, differences in metabolic and ionic

perturbations in 10-20-30 and SIT may explain the different training adaptations observed in the present study.

5 | PERSPECTIVES

The present study shows that intense intermittent training using the 10-20-30 training concept or sprint interval training is easily implemented in-season and is effective to enhance intense intermittent running performance or sprint performance despite a reduction in weekly training time. However, two players from SIT became injured, which unfortunately is common in football.³⁶ Whether the findings of the present study apply to professional players training 6-9 times per week remains to be elucidated. Furthermore, while the Yo-Yo IR1 has been shown to be a good indicator of match performance in football players, future studies should investigate the effect of different intense intermittent training regimes on match performance in elite football players.

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CONFLICT OF INTEREST

Authors have no conflicts related to this work.

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REFERENCES

- Mohr M, Krstrup P, Bangsbo J. Match performance of high-standard soccer players with special reference to development of fatigue. *J Sports Sci*. 2003;21(7):519-528.
- Faude O, Koch T, Meyer T. Straight sprinting is the most frequent action in goal situations in professional football. *J Sports Sci*. 2012;30(7):625-631.
- Rampinini E, Impellizzeri FM, Castagna C, Coutts AJ, Wisloff U. Technical performance during soccer matches of the Italian Serie A league: effect of fatigue and competitive level. *J Sci Med Sport*. 2009;12(1):227-233.
- Iaia FM, Fiorenza M, Perri E, Alberti G, Millet GP, Bangsbo J. The effect of two speed endurance training regimes on performance of soccer players. *PLoS ONE*. 2015;10(9):e0138096.
- Nyberg M, Fiorenza M, Lund A, et al. Adaptations to speed endurance training in highly trained soccer players. *Med Sci Sports Exerc*. 2016;48(7):1355-1364.
- Tønnessen E, Shalfawi SA, Haugen T, Enoksen E. The effect of 40-m repeated sprint training on maximum sprinting speed, repeated sprint speed endurance, vertical jump, and aerobic capacity in young elite male soccer players. *J Strength Cond Res*. 2011;25(9):2364-2370.
- Ingebrigtsen J, Shalfawi SA, Tønnessen E, Krstrup P, Holtermann A. Performance effects of 6 weeks of aerobic production training in junior elite soccer players. *J Strength Cond Res*. 2013;27(7):1861-1867.
- Hostrup M, Bangsbo J. Limitations in intense exercise performance of athletes – effect of speed endurance training on ion handling and fatigue development. *J Physiol*. 2017;595(9):2897-2913.
- Iaia FM, Fiorenza M, Larghi L, Alberti G, Millet GP, Girard O. Short- or long-rest intervals during repeated-sprint training in soccer? *PLoS ONE*. 2017;12(2):e0171462.
- Mohr M, Krstrup P, Nielsen JJ, et al. Effect of two different intense training regimens on skeletal muscle ion transport proteins and fatigue development. *Am J Physiol Regul Integr Comp Physiol*. 2007;292(4):R1594-R1602.
- Ferrari Bravo D, Impellizzeri FM, Rampinini E, Castagna C, Bishop D, Wisloff U. Sprint vs. interval training in football. *Int J Sports Med*. 2008;29(8):668-674.
- Gunnarsson TP, Christensen PM, Holse K, Christiansen D, Bangsbo J. Effect of additional speed endurance training on performance and muscle adaptations. *Med Sci Sports Exerc*. 2012;44(10):1942-1948.
- Gunnarsson TP, Bangsbo J. The 10–20–30 training concept improves performance and health profile in moderately trained runners. *J Appl Physiol*. 2012;113(1):16-24.
- Gliemann L, Gunnarsson TP, Hellsten Y, Bangsbo J. 10–20–30 training increases performance and lowers blood pressure and VEGF in runners. *Scand J Med Sci Sports*. 2015;25(5):e479-e489.
- Toennesen LL, Soerensen ED, Hostrup M, Porsbjerg C, Bangsbo J, Backer V. Feasibility of high-intensity training in asthma. *Eur Clin Respir J*. 2018;5(1):1468714.
- Toennesen LL, Meteran H, Hostrup M, et al. Effects of exercise and diet in nonobese asthma patients-a randomized controlled trial. *J Allergy Clin Immunol Pract*. 2018;6(3):803-811.
- Hostrup M, Onslev J, Jacobson GA, Wilson R, Bangsbo J. Chronic beta2 -adrenoceptor agonist treatment alters muscle proteome and functional adaptations induced by high intensity training in young men. *J Physiol*. 2018;596(2):231-252.
- Christensen PM, Krstrup P, Gunnarsson TP, Küllerich K, Nybo L, Bangsbo J. VO2 kinetics and performance in soccer players after intense training and inactivity. *Med Sci Sports Exerc*. 2011;43(9):1716-1724.
- Thomassen M, Christensen PM, Gunnarsson TP, Nybo L, Bangsbo J. Effect of 2-wk intensified training and inactivity on muscle Na⁺-K⁺ pump expression, phospholemman (FX1D) phosphorylation, and performance in soccer players. *J Appl Physiol*. 2010;108(4):898-905.
- Fransson D, Nielsen TS, Olsson K, et al. Skeletal muscle and performance adaptations to high-intensity training in elite male soccer players: speed endurance runs versus small-sided game training. *Eur J Appl Physiol*. 2018;118(1):111-121.
- Mohr M, Thomassen M, Girard O, Racinais S, Nybo L. Muscle variables of importance for physiological performance in competitive football. *Eur J Appl Physiol*. 2016;116(2):251-262.
- Ortenblad N, Lunde PK, Levin K, Andersen JL, Pedersen PK. Enhanced sarcoplasmic reticulum Ca(2+) release following intermittent sprint training. *Am J Physiol Regul Integr Comp Physiol*. 2000;279(1):R152-160.

23. Bangsbo J, Iaia FM, Krstrup P. The Yo-Yo intermittent recovery test: A useful tool for evaluation of physical performance in intermittent sports. *Sports Medicine*. 2008;38(1):37-51.
24. Krstrup P, Mohr M, Amstrup T, et al. The Yo-Yo intermittent recovery test: physiological response, reliability, and validity. *Med Sci Sports Exerc*. 2003;35(4):697-705.
25. Semenick D. Tests and measurements: the T-test. *Strength Cond J*. 1990;12(1):36-37.
26. Hader K, Mendez-Villanueva A, Palazzi D, Ahmaidi S, Buchheit M. Metabolic power requirement of change of direction speed in young soccer players: not all is what it seems. *PLoS ONE*. 2016;11(3):e0149839.
27. Balsom PD, Seger JY, Sjodin B, Ekblom B. Physiological responses to maximal intensity intermittent exercise. *Eur J Appl Physiol Occup Physiol*. 1992;65(2):144-149.
28. Ortenblad N, Nielsen J, Saltin B, Holmberg HC. Role of glycogen availability in sarcoplasmic reticulum Ca²⁺ kinetics in human skeletal muscle. *J Physiol*. 2011;589(Pt 3):711-725.
29. Hostrup M, Kalsen A, Onslev J, et al. Mechanisms underlying enhancements in muscle force and power output during maximal cycle ergometer exercise induced by chronic ± 2 -adrenergic stimulation in men. *J Appl Physiol*. 2015;119(5):475-486.
30. Vickers AJ, Altman DG. Statistics notes: Analysing controlled trials with baseline and follow up measurements. *BMJ*. 2001;323(7321):1123-1124.
31. Gunnarsson TP, Christensen PM, Thomassen M, Nielsen LR, Bangsbo J. Effect of intensified training on muscle ion kinetics, fatigue development, and repeated short-term performance in endurance-trained cyclists. *Am J Physiol Regul Integr Comp Physiol*. 2013;305(7):R811-R821.
32. Bangsbo J, Gunnarsson TP, Wendell J, Nybo L, Thomassen M. Reduced volume and increased training intensity elevate muscle Na⁺-K⁺ pump α_2 -subunit expression as well as short- and long-term work capacity in humans. *J Appl Physiol*. 2009;107(6):1771-1780.
33. Iaia FM, Thomassen M, Kolding H, et al. Reduced volume but increased training intensity elevates muscle Na⁺-K⁺ pump α_1 -subunit and NHE1 expression as well as short-term work capacity in humans. *Am J Physiol Regul Integr Comp Physiol*. 2008;294(3):R966-R974.
34. Krstrup P, Mohr M, Nybo L, Jensen JM, Nielsen JJ, Bangsbo J. The Yo-Yo IR2 test: physiological response, reliability, and application to elite soccer. *Med Sci Sports Exerc*. 2006;38(9):1666-1673.
35. Seebacher F, Pollard SR, James RS. How well do muscle biomechanics predict whole-animal locomotor performance? The role of Ca²⁺ handling. *J Exp Biol*. 2012;215(Pt 11):1847-1853.
36. Ekstrand J, Gillquist J, Moller M, Oberg B, Liljedahl SO. Incidence of soccer injuries and their relation to training and team success. *Am J Sports Med*. 1983;11(2):63-67.

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